

RESEARCH ARTICLE

Spring phenology rather than climate dominates the trends in peak of growing season in the Northern Hemisphere

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Abstract

Shifts in plant phenology regulate ecosystem structure and function, which feeds back to the climate system. However, drivers for the peak of growing season (POS) in seasonal dynamics of terrestrial ecosystems remain unclear. Here, spatial-temporal patterns of POS dynamics were analyzed by solar-induced chlorophyll fluorescence (SIF) and vegetation index in the Northern Hemisphere over the past two decades from 2001 to 2020. Overall, a slow advanced POS was observed in the Northern Hemisphere, while a delayed POS distributed mainly in northeastern North America. Trends of POS were driven by the start of growing season (SOS) rather than pre-POS climate both at hemisphere and biome scale. The effect of SOS on the trends in POS was the strongest in shrublands while the weakest in evergreen broad-leaved forest. These findings highlight the crucial role of biological rhythms rather than climatic factors in exploring seasonal carbon dynamics and global carbon balance.

KEYWORDS

biological rhythms, climate change, end of growing season, gross primary productivity, solar-induced chlorophyll fluorescence, start of growing season

1 | INTRODUCTION

Land surface phenology responds to climate change and affects global carbon cycle and energy balance (Penuelas et al., 2009; Piao et al., 2020). The peak of growing season (POS), which indicates the time when vegetation photosynthetic capacity reaches its maximum, plays an important role in characterizing the capacity of terrestrial ecosystem productivity and shaping gross primary productivity (GPP) dynamics during the growing season (Huang et al., 2018; Park et al., 2019). Maximum productivity at POS explains about 78% of the variation of seasonal total productivity (Xia et al., 2015). Differential shift in POS across biomes means keen competition for resource availability, profoundly affecting ecosystem structure and evolution (Radville et al., 2016; Steltzer & Post, 2009). Whether the Northern Hemisphere POS advance or not still appears to be a matter of debate under global warming. An overall advance in POS has been found

across extratropical Northern Hemisphere through 34-year remote-sensing normalized difference vegetation index (NDVI) observations (Gonsamo et al., 2018). However, recent research suggested that no advanced trend of POS has been observed in Northern Hemisphere mid-latitudes whether derived from GPP (Ge et al., 2022), or derived from NDVI (Zhao et al., 2022). Therefore, trends in POS need to be further explored in the Northern Hemisphere under climate change.

Climate is considered as the main drivers of POS variation (Liu et al., 2021; Wang & Wu, 2019). Previous studies showed that temperature was the major driver of POS advance both in the United States and in China (Liu et al., 2021; Wang & Wu, 2019). Xu et al. (2016) found that warming drove the advance of POS by increasing the growing-degree days accumulation in a year in the mid-latitudes of the Northern Hemisphere (Xu et al., 2016). In recent years, the non-negligible effect of biological rhythms on phenology has been proved (Keenan & Richardson, 2015). Liu

et al. (2016) found that delayed autumn phenology connected with earlier spring phenology for some biomes (i.e., DBF, MF) in the Northern Hemisphere (Liu et al., 2016). Shen et al. (2020) detected a strongly positive correlation between the previous year's end of growing season (EOS) and start of growing season (SOS) in the boreal region and a weak negative correlation in temperate ecosystems (Shen et al., 2020). The results from field control experiments and long-term phenology network observations indicated that variation in leaf flushing date influenced leaf senescence as well as flushing date in the coming year (Fu et al., 2014). The relative importance of biological rhythms and climate for POS trends remains unclear. Therefore, detangling the relative contribution of climate versus biological rhythms is essential for understanding POS changes in the Northern Hemisphere.

SIF derived from remote-sensing observations provide new insights to monitor land surface phenology, particularly at regional and global scale (Jeong et al., 2017). Chlorophyll fluorescence (CF), as the re-emission of part of the energy absorbed by chlorophyll during plant photosynthesis, provides a distinctive "glow" of photosynthetically active vegetation (Baker, 2008; Jeong et al., 2017). SIF is directly linked to photosynthetic activity and insensitive to atmospheric scattering, cloud and snow cover (Chang et al., 2019; Sun et al., 2018), which can be used to observe physiological phenology at multiple scale from leaf to global (Jeong et al., 2017). SIF-based land surface phenology studies have shown remarkable advantages of detecting subtle phenological changes compared to traditional vegetation indicator (Chen et al., 2022; Zhou et al., 2022). For example, compared with enhanced vegetation index (EVI) and NDVI, SIF can track SOS and EOS of subtropical evergreen coniferous forests perfectly (Zhou et al., 2020). Furthermore, inversion of SOS and EOS based on SIF confirms to the actual physiological conditions of vegetation, rather than the VIs-based phenology (Yang et al., 2019). NDVI was frequently used for POS monitoring, but recent studies found that NDVI can only track peak greenness rather than peak photosynthesis (Ge et al., 2022). However, less studies used SIF to track POS even if the machine learning method breaks the limitation of remote-sensing SIF time span (Zhang et al., 2018).

The Northern Hemisphere occupies the main part of the terrestrial ecosystems, including diverse plant functional groups and plant species. The extratropical ecosystems of the Northern Hemisphere are experiencing epidemic warming, drought, regional imbalance of water resources, and increasingly frequent weather extreme (Coumou & Rahmstorf, 2012). In this study, we estimated the POS derived from contiguous solar-induced fluorescence (CSIF) and vegetation index (NDVI) dataset for the areas with significant seasonal dynamics in vegetation activities in the Northern Hemisphere ($>30^{\circ}\text{N}$) from 2001 to 2020. The primary objects of studies were (1) to compare the ability of tracking POS derived from SIF and NDVI, (2) to explore the spatial and temporal patterns of POS in the Northern Hemisphere, and (3) to investigate the relative importance of climate and biological rhythms for POS trends.

2 | MATERIALS AND METHODS

2.1 | Dataset

2.1.1 | SIF dataset

CSIF dataset from 2001 to 2020 was employed in current study, which was derived from *Orbiting Carbon Observatory 2* (OCO-2) SIF and Moderate Resolution Imaging Spectroradiometer (MODIS) reflectance dataset (MCD43C4 V006) using machine learning algorithms, including clear-sky SIF sub-dataset after eliminating cloud and aerosol interference, with a spatial resolution of $0.5^{\circ}\times 0.5^{\circ}$ and a 4-day temporal resolution (<https://figshare.com/articles/dataset/CSIF/6387494>) (Zhang et al., 2018). Negative values of clear-sky SIF dataset were replaced with 0. Pixels with SIF more than 10 missing data in a year were excluded.

2.1.2 | NDVI dataset

NDVI dataset during 2001–2020 was from MODIS vegetation indices product (MOD13C1 V006). MOD13C1 data were cloud-free spatial composites of the MOD13A2 data, with a spatial resolution of $0.05^{\circ}\times 0.05^{\circ}$ and a 16-day temporal resolution (<https://lpdaac.usgs.gov/products/mod13c1v006/>). The spatial resolution of the NDVI dataset was aggregated to 0.5° using the mean method to match the SIF dataset.

2.1.3 | Climate dataset

Monthly grid datasets for temperature and precipitation during 2001–2020 were obtained from CRU-TS 4.05 (<https://www.uea.ac.uk/web/groups-and-centres/climatic-research-unit/data>), with a spatial resolution of $0.5^{\circ}\times 0.5^{\circ}$. The datasets were generated by interpolating monthly climatic anomalies from meteorological station observation networks (Harris et al., 2020) and have been widely used in recent vegetation phenology studies (Peng et al., 2013; Zeng et al., 2021). Monthly grid dataset for downward short-wave radiation with a spatial resolution of $0.25^{\circ}\times 0.25^{\circ}$ during 2001–2020 was obtained from NASA Global Land Data Assimilation System Version 2.1 (GLDAS-2.1), which was forced with a combination of observation data and model since 2000 (https://disc.gsfc.nasa.gov/datasets/GLDAS_NOAH025_M_2.1/summary) (Rodell et al., 2004). We aggregated the downward short-wave radiation dataset to $0.5^{\circ}\times 0.5^{\circ}$ spatial resolution.

2.1.4 | Land cover dataset

MODIS land cover type MCD12C1 dataset with International Geosphere-Biosphere Programme (IGBP) land cover classification scheme was applied to identify vegetation types, which was

derived through a supervised decision-tree classification method, with a spatial resolution of $0.05^\circ \times 0.05^\circ$ and consist of 17 land cover classes (<https://lpdaac.usgs.gov/products/mcd12c1v006>) (Friedl et al., 2010). In all, 10 major vegetation types including evergreen needleleaf forest (ENF), evergreen broadleaf forest (EBF), deciduous needleleaf forest (DNF), deciduous broadleaf forest (DBF), mixed forest (MF), closed shrublands (CSH), open shrublands (OSH), woody savannas (WSAV), savannas (SAV), and grasslands (GRA) were considered in this study. Moreover, we aggregated it to $0.5^\circ \times 0.5^\circ$ using the majority rule to match the spatial resolution of SIF, NDVI data, and climate variables (Figure S1). Without considering landcover change, we used a medium year (2011) to characterize biomes in the Northern Hemisphere.

2.2 | Phenology extraction methods

The area covered with barren or sparse vegetation was first excluded from our study by the criteria of maximum SIF values <0.1 or annual mean NDVI values lower than 0.1 (Wang et al., 2020). Moreover, pixels dominated with urban or cropland were excluded according land cover classification to minimize the impact of human activities (Liu et al., 2016). The phenology retrieval is affected by the extraction methods. Thus, we used three methods to evaluate SOS and POS, and averaged the phenological metrics from three methods.

2.2.1 | Double logistic derivation method

We first used Savitzky-Golay filter to SIF/NDVI time series, for filling the gaps and minimizing the noise before retrieving phenology metrics (Wang et al., 2019). Then a double logistic growth model (Equation 1) was conducted to fit the data times series and to reconstruct curve of annual SIF and NDVI (Gonsamo et al., 2013). As a seven-parameter logistic function, it provides a robust and flexible way in capturing the seasonal dynamics of vegetation. Two dates of the two local maxima points in second-order derivative of the fitted curve were corresponding to the date of SOS and EOS. POS was defined as the date when the value reaches maximum in the fitted SIF/NDVI curve according to previous study (Gonsamo et al., 2013).

$$f(x) = \alpha_1 + \frac{\alpha_2}{1 + e^{-\partial_1(x-\beta_1)}} - \frac{\alpha_3}{1 + e^{-\partial_2(x-\beta_2)}} \quad (1)$$

where x is the day of year (DOY), α_1 is the background values, α_2 - α_1 is the difference between the background and the amplitude of the spring and early summer plateau, and α_3 - α_1 is the difference between the background and the amplitude of the late summer and autumn plateau both in SIF/NDVI units. ∂_1 and ∂_2 are the transition curvature parameters, and β_1 and β_2 are the midpoints in DOYs of these transitions for greenup and browningdown, respectively (Gonsamo et al., 2013).

2.2.2 | Polynomial fit maximum rate method

A six-order polynomial (Equation 2) was applied to fit SIF/NDVI time series and to predict values of daily SIF/NDVI in a year, which was a common method for fitting seasonal dynamics curve of vegetation (Piao et al., 2006). Subsequently, SOS was extracted through a fixed threshold, the threshold is defined as the SIF/NDVI value corresponding to the date (step for SIF/NDVI is 4 or 16 days, respectively) before the occurrence of the maximum change rate in the 20-year average SIF/NDVI time series (Equation 3). The determination of POS was same as the double logistic derivation method.

$$f(x) = a + a_1x^1 + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 \quad (2)$$

where x is the day of year (DOY), and $a, a_1, a_2, a_3, a_4, a_5, a_6$ are the fitted equation coefficients.

$$f_{ratio}(t) = [y(t+1) - y(t)] / y(t) \quad (3)$$

where t is the observed date (with 4 or 16 days temporal resolution for SIF/NDVI, respectively).

2.2.3 | Spline threshold method

A cubic smoothing spline model (Equation 4) was used to fit SIF/NDVI time series and to predict values of daily SIF/NDVI in a year, which was flexible to fit seasonal variation patterns of vegetation and effective in reconstructing seasonal trajectories (Chen et al., 2006). SOS was evaluated as the date when dynamic threshold (the ratio of the observed value minus the annual minimum value to the annual maximum value minus the minimum value) reaches 0.2 (Equation 5) (White et al., 1997). The determination of POS was same as the double logistic derivation method.

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 \quad (4)$$

where x and x_i are the adjacent two points (step for SIF/NDVI is 4 or 16 days, respectively). a_i, b_i, c_i , and d_i are the fitted coefficients of equation. The SIF/NDVI values between each adjacent two points was evaluated from several cubic polynomial S_i .

$$f_{ratio}(t) = [y(t) - y(t)_{\min}] / [y(t)_{\max} - y(t)_{\min}] \quad (5)$$

where t is the time for reconstructed SIF/NDVI time series, $y(t)$ is SIF/NDVI values at the date of t , $y(t)_{\min}$ and $y(t)_{\max}$ are the annual minimum SIF/NDVI values and maximum SIF/NDVI values, respectively.

2.3 | Analyses

Temporal trends of SOS, POS, and pre-POS meteorological variables were calculated by the Theil-Sen nonparametric slope

estimator. A nonparametric Mann–Kendall test was applied to evaluate the significance of these trends (Gocic & Trajkovic, 2013). The advantages of Mann–Kendall method allow missing values and need not for the data to conform to any particular distribution, and the feature of Theil–Sen method is that slope estimator is not affected by outliers (Gilbert, 1987). Hence, two methods were widely used to trend analysis for time-ordered date (Park et al., 2019; Wang et al., 2019). Statistically significant was considered when probability value <0.05. To ensure robust analysis results, statistical analysis is only performed on pixels without missing phenology values for 20 years.

To further investigate how the climate and biological rhythms affect POS, spearman partial correlation analysis was applied to detect the correlation between POS and each driver (i.e., pre-POS temperature, pre-POS precipitation, pre-POS radiation, and SOS) while controlling the influence of other factors. As vegetation phenology is more affected by pre-season climate, it is usually used pre-season climate to analyze the response of vegetation phenology to climatic factors (Piao et al., 2015; Wang & Wu, 2019), we first calculated the pre-POS length for each driver. The pre-POS length for temperature was determined as the period preceding the multiyear average of POS (in steps of 1 month) when the largest partial correlation coefficient has occurred between POS and temperature after eliminating the influence of precipitation, radiation over the same period and SOS, multiyear average month of POS to the multiyear average month of SOS was set as the maximum range of this period. Similar steps were performed for precipitation and radiation. When the pre-POS length of temperature, precipitation, and radiation was estimated, the impact of SOS on POS can be accessed by partial correlation coefficient after controlling pre-POS temperature, precipitation, and radiation. Similar steps were performed for temperature, precipitation, and radiation. Moreover, partial correlation coefficient between four variables and POS across biomes was calculated by averaged values from all pixels in each biome, because the effects of climate and biological rhythms on phenology varied among vegetation types.

A partial derivative-based approach was applied to quantify the relative importance of biological rhythms and climatic factors on POS dynamics, which has been widely employed to assess the impact of climatic and non-climatic factors on hydrology (Meng & Mo, 2012), evapotranspiration dynamics (Li et al., 2017; Yang & Yang, 2012), vegetation productivity (Yan et al., 2019), and phenology (Yuan et al., 2020) in recent years. We assumed that the POS trends were influenced by climatic factors, biological rhythms, and other factors. Thus, the trends of POS (S_{POS}) can be described as:

$$\begin{aligned} S_{POS} &= Con_{-climate} + Con_{-bio-rhythms} + Con_{-other} \\ &= Con_{temp} + Con_{prec} + Con_{radi} + Con_{SOS} + Con_{other} \\ &= \frac{\partial POS}{\partial temp} \times \frac{d_{temp}}{d_t} + \frac{\partial POS}{\partial prec} \times \frac{d_{prec}}{d_t} + \frac{\partial POS}{\partial radi} \times \frac{d_{radi}}{d_t} + \frac{\partial POS}{\partial SOS} \times \frac{d_{SOS}}{d_t} + Con_{other} \end{aligned} \quad (6)$$

where $Con_{-climate}$, $Con_{-bio-rhythms}$, and Con_{-other} represent the contributions of climatic factors, biological rhythms, and other factors to POS trends,

respectively. A positive value represents a contribution to the delay of POS, while a negative value represents a contribution to the advance of POS. $Con_{-climate}$ is the sum of Con_{temp} , Con_{prec} , Con_{radi} , and Con_{SOS} . Con_{SOS} is equal to $Con_{temp} \times \frac{\partial POS}{\partial temp} + Con_{prec} \times \frac{\partial POS}{\partial prec} + Con_{radi} \times \frac{\partial POS}{\partial radi}$, and $\frac{\partial POS}{\partial temp}$, $\frac{\partial POS}{\partial prec}$, $\frac{\partial POS}{\partial radi}$, and $\frac{\partial POS}{\partial SOS}$ represent sensitivities of POS to temperature (days/°C), precipitation (days/mm), radiation (days/(W/m²)), and SOS (days/day), respectively, which can be computed as the linear slope by multiple linear regression analysis. $\frac{d_{temp}}{d_t}$, $\frac{d_{prec}}{d_t}$, $\frac{d_{radi}}{d_t}$, and $\frac{d_{SOS}}{d_t}$ represent the inter-annual variation rate of temperature, precipitation, radiation, and SOS, respectively, which can be accessed by Theil–Sen slope estimator.

Subsequently, without considering other factors, the rate of climatic factors and biological rhythms for POS dynamics can be estimated through contribution proportions:

$$Rate_{(SOS,temp,prec,radi)} = \frac{Con_{(SOS,temp,prec,radi)}}{Con_{SOS} + Con_{temp} + Con_{prec} + Con_{radi}} \quad (7)$$

where $Con_{(SOS,temp,prec,radi)}$ represents the contributions of each variable for POS trends; $Rate_{(SOS,temp,prec,radi)}$ represents the contribution proportions of each variable. A positive value represents the same direction between contributions of drivers and trends of POS, while a negative value represents an opposite direction.

3 | RESULTS

3.1 | Spatial and temporal patterns of POS

Mean POS_{CSIF} in the Northern Hemisphere during the period 2001–2020 ranged from day 179 at relatively low latitudes (approximately 38°N) to day 210 in the high latitudes, with POS_{CSIF} in most pixels ranged from day 180 to day 200 (Figure 1a). In detail, minimum POS_{CSIF} occurred in approximately 38°N, then increase along the latitude on both sides. POS_{NDVI} showed a similar latitudinal pattern to POS_{CSIF}, but POS_{NDVI} occurred later than POS_{CSIF} (Figure 1c). Mean SOS_{CSIF} and SOS_{NDVI} exhibited similar latitudinal and spatial pattern, while an earlier SOS_{NDVI} can be obviously observed than SOS_{CSIF} in the North Hemisphere (Figure S2a,c). Although the trends of POS_{CSIF} and POS_{NDVI} had similar spatial distribution, the specific proportions were different. Advanced trend of POS_{CSIF} occurred in about 65.09% of the study area in the Northern Hemisphere during 2001–2020, but this proportion of advanced trend for POS_{NDVI} was 53.41% (Figure 1b,d). Statistically significant advanced area ($p < .05$) primarily distributed in central Siberia and eastern Asia, while maximal advance of POS occurred in eastern China. In contrast, only 27.88% of the study area showed delayed POS_{CSIF} (39.71% of the study area experienced delayed POS_{NDVI}), mostly in northeastern North America and western Siberia (Figure 1b,d). SOS (including SOS_{CSIF} and SOS_{NDVI}) trends showed a similar spatial pattern to POS trends although SOS experienced more areas of advanced trends (74.89% for SOS_{CSIF}, 74.47% for SOS_{NDVI}) and less areas of delayed trends (21.76% for SOS_{CSIF}, 22.97% for SOS_{NDVI}) compared to POS (Figure S2b,d).

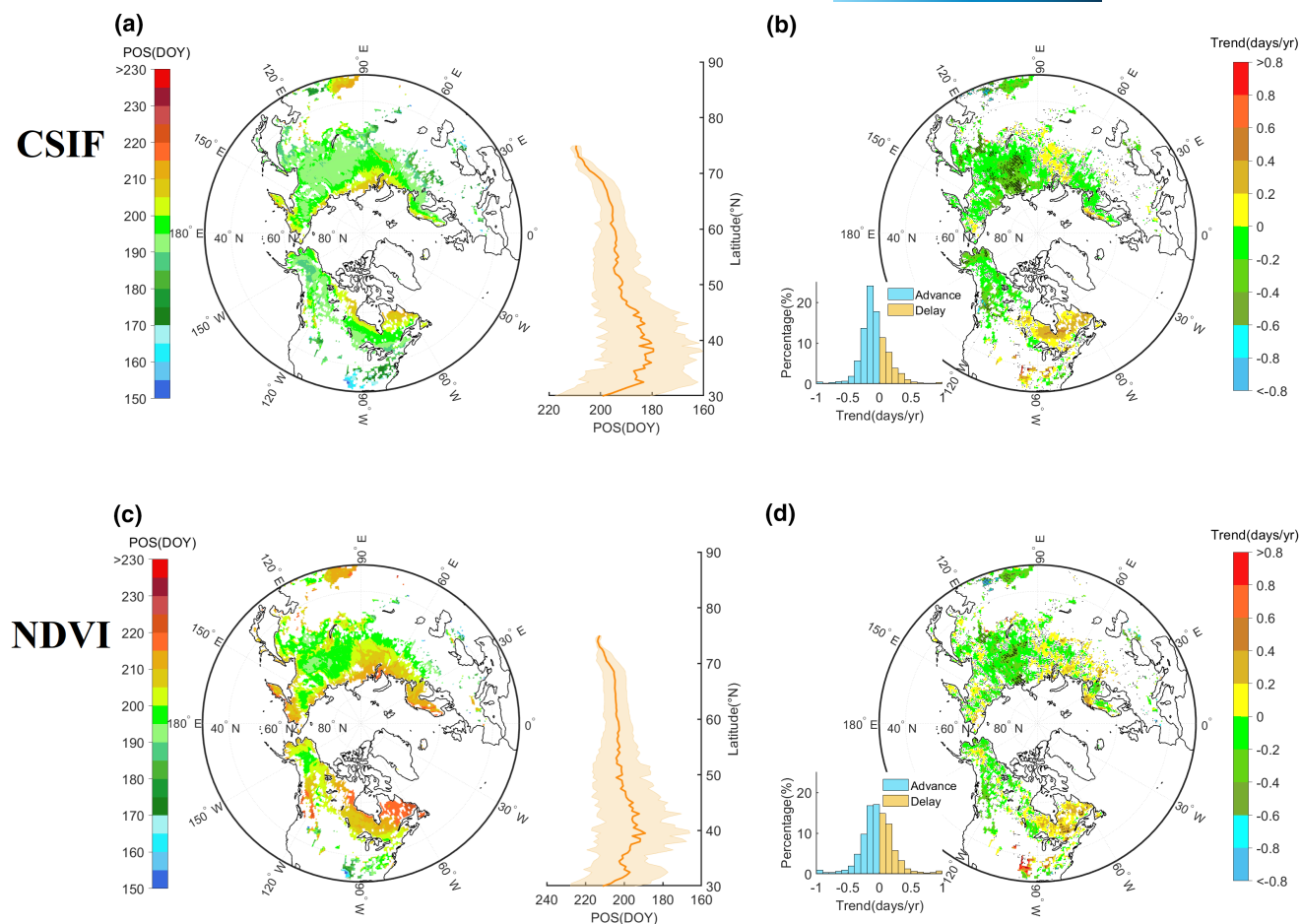
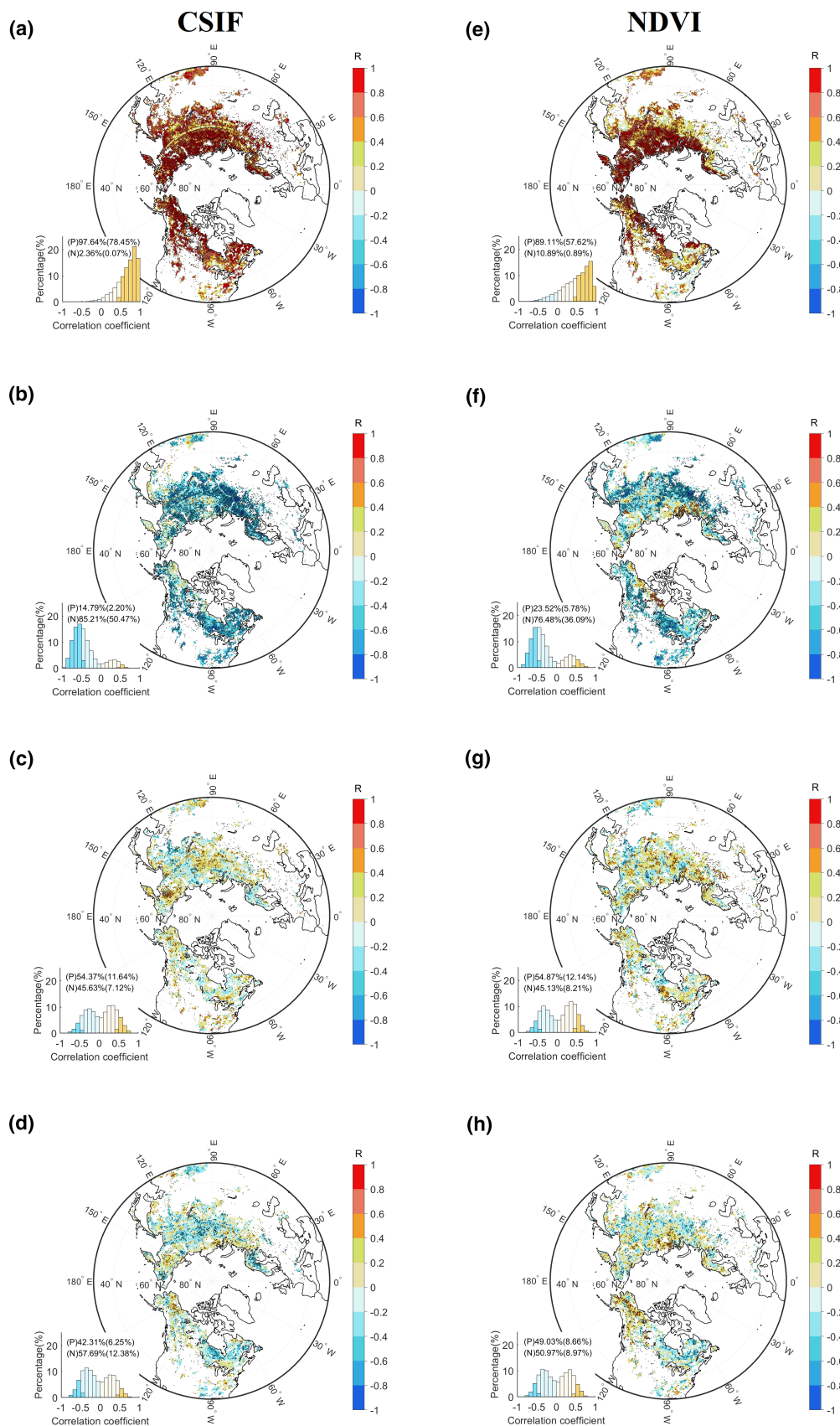


FIGURE 1 (a, c) Spatial distribution of the average POS (determined by the average of three extraction methods) over the North Hemisphere ($>30^{\circ}\text{N}$) during 2001–2020, the right axis showed the average POS estimated by the ensemble mean of each latitude. (b, d) Spatial distribution of POS trends over the North Hemisphere ($>30^{\circ}\text{N}$) during 2001–2020, the lower left histogram in (b, d) showed the proportion of POS trends at different slopes, the dots on the colorful regions indicated the detected trends were significant ($p < .05$), the dots on the blank regions indicated that no trend has been detected (trend = 0). Blank pixels were excluded. Subscripts CSIF and NDVI denoted the data sources. CSIF, contiguous solar-induced fluorescence; NDVI, normalized difference vegetation index; POS, peak of growing season. [Colour figure can be viewed at wileyonlinelibrary.com]

3.2 | Relationships of POS with climate and biological rhythms in the Northern Hemisphere

Maximum proportions of pre-POS length of temperature occurred in 2 months for POS_{CSIF} and POS_{NDVI} , while pre-POS length of precipitation and radiation occurred mainly in 0, 1, and 2 months for POS_{CSIF} and occurred mainly in 0, 1, 2, and 3 months for POS_{NDVI} (Figure S3). For most study area experienced increasing pre-POS temperature (62.86% for POS_{CSIF} and 61.37% for POS_{NDVI}) and pre-POS radiation (74.25% for POS_{CSIF} , 69.94% for POS_{NDVI}), while the areas with increased pre-POS precipitation (51.82% for POS_{CSIF} , 52.64% for POS_{NDVI}) were slightly larger than areas with reduced precipitation (42.58% for POS_{CSIF} , 42.08% for POS_{NDVI}) (Figure S4). Partial correlation analysis of POS_{CSIF} and POS_{NDVI} with factors exhibited approximately the same spatial distribution (Figure 2). Positive correlations between POS_{CSIF} and SOS occurred in 97.64% of the study area (89.11% for POS_{NDVI}), with significant ($p < .05$) positive correlations in approximately 78.45% of the study area (57.62%

for POS_{NDVI}). Only 2.36% of the study area showed negative correlations between POS_{CSIF} and SOS (10.89% for POS_{NDVI}) (Figure 2a,e). Similar results of strong positive correlations also confirmed by simple correlation analysis as well as partial correlation analysis after controlling cumulative temperature, precipitation, and radiation (Figure S5). Temperature exhibited negative correlations with POS for most areas (85.21% for POS_{CSIF} , 76.48% for POS_{NDVI}), with statistically significant ($p < .05$) in roughly 50.47% and 36.09% of the study area for POS_{CSIF} and POS_{NDVI} , while positive correlations between POS_{CSIF} and temperature observed in 14.79% (23.52% for POS_{NDVI}) of regions such as eastern Europe and eastern Asia (Figure 2b,f). Meanwhile, more than half regions showed positive correlations between POS and precipitation (54.37% for POS_{CSIF} , 54.87% for POS_{NDVI}) (Figure 2c,g). About 57.69% of the study area exhibited positive correlations between POS_{CSIF} and radiation, around 12.38% were significant ($p < .05$), distributing sparsely in Europe and northwestern North America, while negative correlations detected in 42.31% of the study area, with statistically significant ($p < .05$)

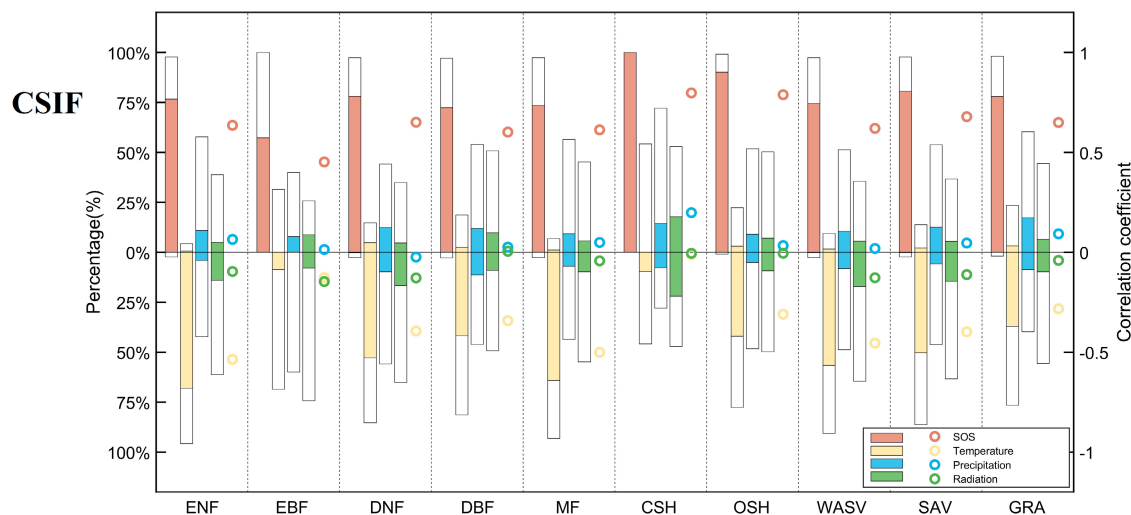


observed in 6.25% of the study area (Figure 2d). However, neither the positive (50.97%) nor the negative (49.03%) correlations between POS_{NDVI} and radiation dominated the study area (Figure 2h).

The relationships between SOS, climatic factors, and POS varied among biomes (Figure 3). In general, different proportions of significantly positive correlations between POS and SOS among all

FIGURE 2 Spatial distribution of partial correlation coefficient between POS and (a, e) SOS, (b, f) temperature, (c, g) precipitation and (d, h) radiation in the North Hemisphere ($>30^{\circ}\text{N}$) during 2001–2020. The lower left histogram showed the proportion of the partial correlation coefficient between POS and corresponding factor for different intervals, with brighter parts indicated that the correlations were significant ($p < .05$). The text in histogram indicated the percentage of areas of positive correlation (P) and negative correlation (N) between POS and corresponding factor, with significant percentages in parentheses. The dots on the colorful regions indicated the detected correlations were significant ($p < .05$). Blank pixels were excluded. Subscripts CSIF and NDVI denoted the data sources. CSIF, contiguous solar-induced fluorescence; NDVI, normalized difference vegetation index; POS, peak of growing season; SOS, start of growing season. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

(a)



(b)

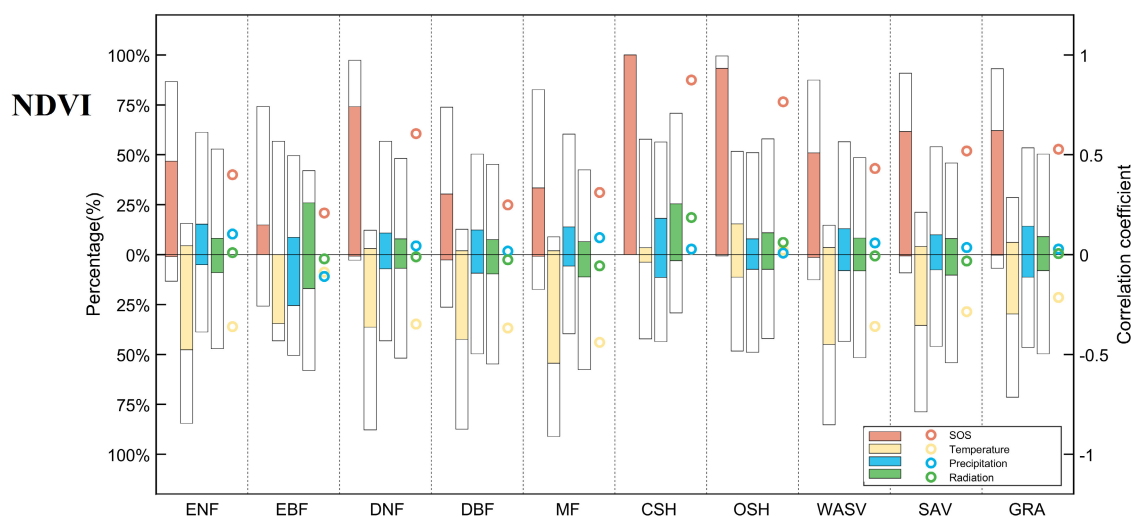


FIGURE 3 Partial correlation coefficient between (a) POS_{CSIF} , SOS and climatic variables (temperature, precipitation and radiation) of each biome (statistic across pixels with the same biome type), and partial correlation coefficient between (b) POS_{NDVI} , SOS and climatic variables of each biome. Bars above 0 represented percentages of positive correlations, while the underneath showed negative percentages. Colored parts indicated proportion of significant correlations ($p < .05$). Mean values of the partial correlation coefficient was also provided (colorful dots). Subscripts CSIF and NDVI denoted the data sources. CSIF, contiguous solar-induced fluorescence; NDVI, normalized difference vegetation index; POS, peak of growing season; SOS, start of growing season. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

biomes, which was the most dominant in CSH (around 100% and 100% of the areas for POS_{CSIF} and POS_{NDVI} , respectively) and OSH (around 99.09% and 99.42% of the areas for POS_{CSIF} and POS_{NDVI} , respectively), with the highest mean correlation coefficient in CSH (0.80 and 0.87 for POS_{CSIF} and POS_{NDVI} , respectively) and OSH

(0.79 and 0.77 for POS_{CSIF} and POS_{NDVI} , respectively). The significantly positive correlations between POS_{CSIF} and SOS were stronger in coniferous forests (76.66% and 78.03% of the areas for ENF and DNF, respectively) than that in broad-leaved forests (57.28% and 72.38% of the areas for EBF and DBF, respectively), a similar

result proved by POS_{NDVI} . Temperature was negatively associated with POS for all biomes, with large proportions of significantly negative correlations in ENF (68.00% and 47.61% of the areas for POS_{CSIF} and POS_{NDVI} , respectively) and MF (64.15% and 54.44% of the areas for POS_{CSIF} and POS_{NDVI} , respectively), with the highest mean correlation coefficient in ENF (-0.54 , -0.36 for POS_{CSIF} and POS_{NDVI} , respectively) and MF (-0.50 , -0.44 for POS_{CSIF} and POS_{NDVI} , respectively) (Figure 3). Compared to temperature, precipitation showed a weaker and inconsistent relationship with POS among biomes. For radiation, the areas of negative correlations with POS were larger than that of positive correlations in most biomes, except for CSH and OSH.

3.3 | Contributions of climate and biological rhythms to trends in POS in the Northern Hemisphere

In general, the contributions of influential factors (i.e., SOS, temperature, precipitation, and radiation) showed spatial heterogeneity (Figure 4). SOS contributed to an advanced POS can be detected in most of the study area, a large contribution was found in eastern Asia and central Siberia, whereas SOS changes were linked to a delayed POS in northern Europe and northeastern North America (Figure 4a,e). Temperature contributed to POS advance for most study area, with large values in high latitude such as northwestern North America (Figure 4b,f). Compared to SOS and temperature, precipitation and radiation showed weaker contributions to POS trends (Figure 4c,d,g,h).

Overall, the contribution proportions of SOS to POS trends were the highest, with roughly 51.10% and 41.45% for POS_{CSIF} and POS_{NDVI} , respectively (Figure 5). SOS showed high contribution proportions to POS trends in most regions of the study area. However, low contribution proportions occurred in east North America, east Asia, and central Siberia (Figure S6a,f). The contribution proportions of temperature were 30.47% and 29.56% for POS_{CSIF} and POS_{NDVI} , respectively (Figure 5). Pervasive warming produced different trends for POS. High temperature contribution proportions induced by rapid warming were observed in central Siberia, but temperature hindered ($<0\%$) POS dynamics in some region such as east Asia (Figure S6b,g; Figure S4a,d). The contribution proportions of precipitation were lowest, accounting for 7.98% and 14.13% for POS_{CSIF} and POS_{NDVI} , respectively (Figure 5). Moreover, the contribution proportions of radiation were slightly larger than that of precipitation, about 10.45% and 14.86% for POS_{CSIF} and POS_{NDVI} , respectively.

In all 10 vegetation biomes, SOS was considered as the most important driver for POS dynamics in most biomes of the Northern Hemisphere, especially in CSH (63.74% for POS_{CSIF} , 61.36% for POS_{NDVI}) and OSH (57.98% for POS_{CSIF} , 62.23% for POS_{NDVI}) (Figure 5). Meanwhile, temperature was the most essential climatic factor for POS dynamics, except for EBF and CSH (Figure 5). Notably, the dominant factors of POS_{CSIF} and POS_{NDVI} were different in some biomes (such as ENF, DBF, and MF), indicating that SOS was the

dominant factor of POS_{CSIF} while temperature-driven changes in POS_{NDVI} . In addition, precipitation and radiation play a weaker role in POS dynamics than that of temperature and SOS in most biomes.

4 | DISCUSSION

4.1 | A slow advance of POS in the Northern Hemisphere

A slow advanced trend of POS with a rate of 0.83 days per decade for POS_{CSIF} and 0.25 days per decade for POS_{NDVI} was found in the Northern Hemisphere ($>30^\circ\text{N}$) from 2001 to 2020 based on three phenological extraction methods. The trend of POS was consistent with previous studies. For instance, Park et al. (2019) detected a significantly earlier peak photosynthetic with a rate of 1.66 ± 0.30 days per decade across northern lands ($>30^\circ\text{N}$) through 17-year time series of MODIS GPP (Park et al., 2019). Using 34-year GIMMS (Global Inventory Modeling and Mapping Studies) NDVI data, Gonsamo et al. (2018) found that POS was shifting toward spring by about 1.2 ± 0.6 days per decade across extratropical ($>23^\circ\text{N}$) region (Gonsamo et al., 2018). However, the magnitude of advanced POS varied with different studies, which could be related to difference in study periods, methodology, and data sources.

4.2 | Dominant role of SOS rather than climate in POS trends

SOS rather than the pre-POS climatic factors was found to be the most important driver for POS variations in the Northern Hemisphere and among most biomes (Figure 5). Previous studies, however, provided strong evidence of temperature controlled on POS trends. Park et al. (2019) confirmed a widespread warming induced advance in POS in the Northern hemisphere (Park et al., 2019). Furthermore, Wang et al. (2019) and Liu et al. (2021) found that temperature was the most related factor for POS in China and America, respectively (Y. Liu et al., 2021; Wang & Wu, 2019). Nevertheless, our results confirmed the views that what happens later was contingent on what happened earlier for plant life cycle (Donohue, 2009). First, an endogenous memory effect (or biological carryover effects) of plant may play an important role (Ogle et al., 2015). Recent studies have found that spring vegetation growth exerted strong biological carryover effects on summer vegetation growth over climatic factors. This dominant effect of previous stage for later stage may explain the underlying mechanism of SOS dominating trends of POS due to the tight relationship between growth and development (Lian et al., 2021; Y. Yuan et al., 2022). Moreover, conservative strategy (or higher phenological plasticity) of plants to ongoing climate change should be considered. For instance, plants adapted to constant warming by reducing temperature sensitivity of phenology to avoid mortality under high temperatures (Allen et al., 2010; Fu et al., 2015).

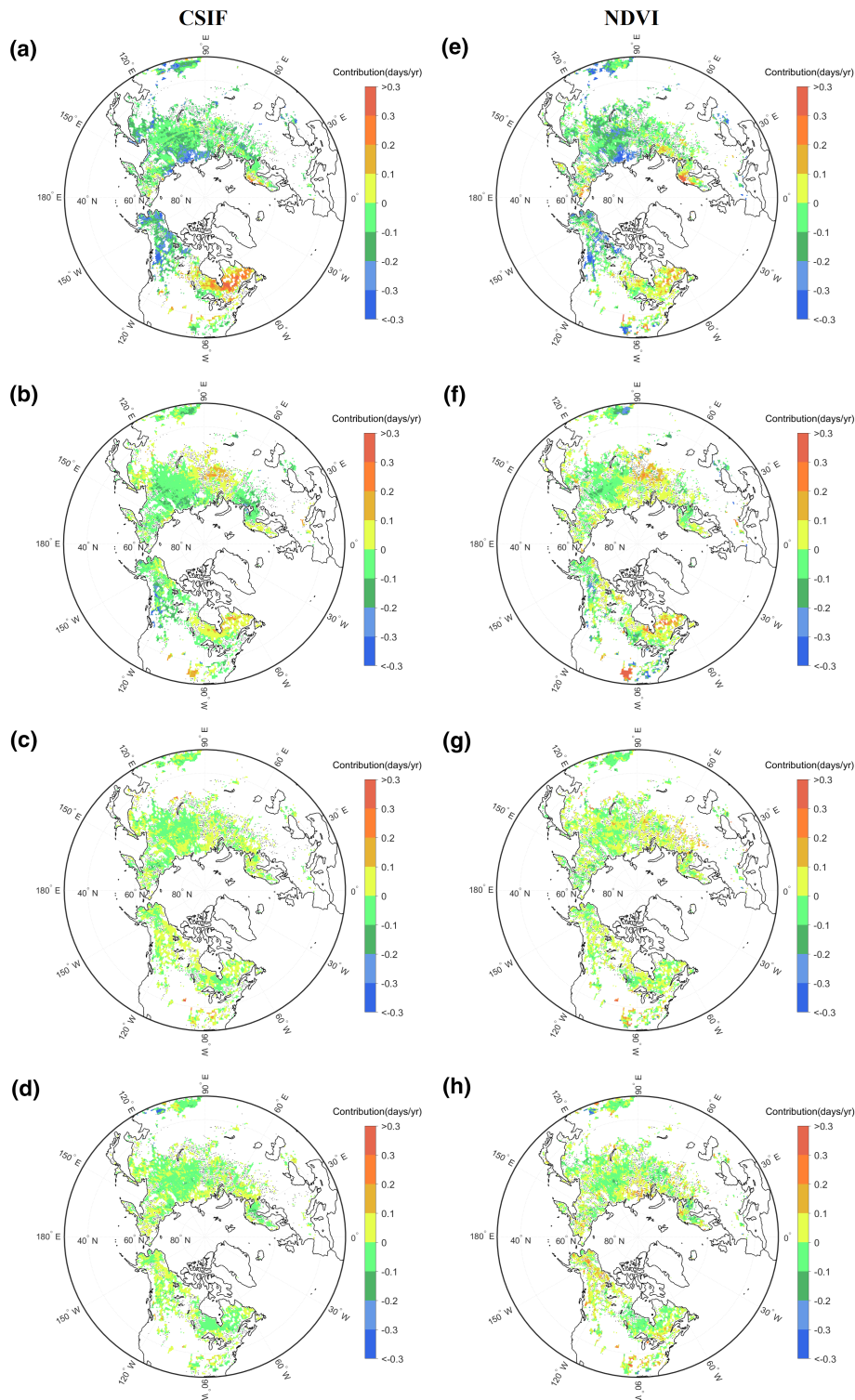
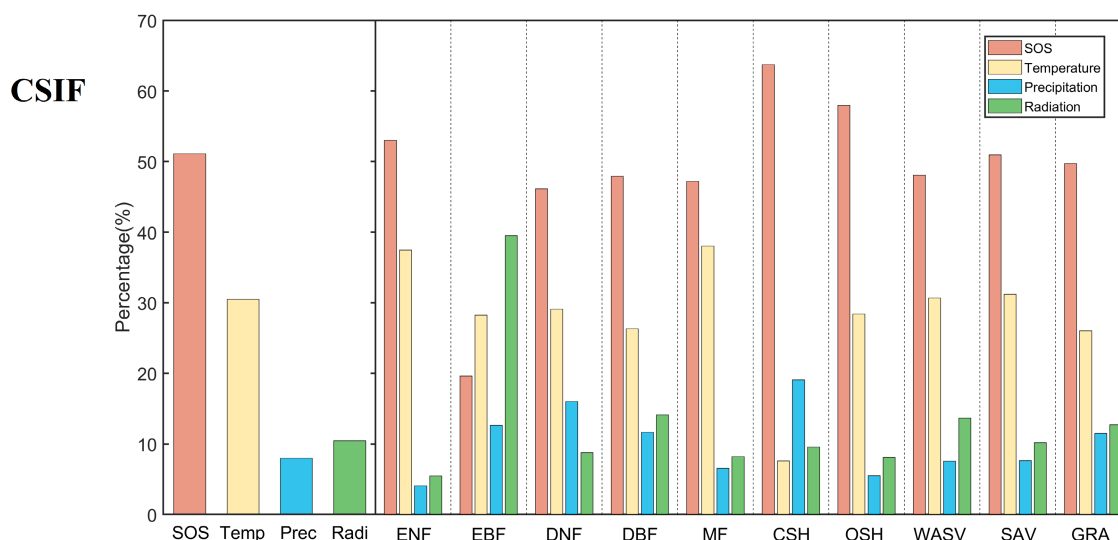


FIGURE 4 Spatial distribution of the contributions of (a, e) SOS, (b, f) temperature, (c, g) precipitation, and (d, h) radiation to POS variations over the North Hemisphere ($>30^{\circ}\text{N}$) during the period from 2001 to 2020. The dots on the blank regions indicated that no trend has been detected (trend = 0). Blank pixels were excluded. Subscripts CSIF and NDVI denoted the data sources. CSIF, contiguous solar-induced fluorescence; NDVI, normalized difference vegetation index; POS, peak of growing season; SOS, start of growing season. [Colour figure can be viewed at wileyonlinelibrary.com]

In addition, the leading role of SOS is differential among biomes (Figure 5). Specifically, the dominant role of SOS in shrublands (both CSH and OSH) was stronger than that in other biomes. This suggested that biome-specific rhythm should be considered in exploring

the trends in POS in the Northern Hemisphere. We proposed two hypotheses for this interesting result: (1) discrepantly ecological memory was strongest in shrublands among biomes and (2) differential response to climate change among biomes existed. For the

(a)



(b)

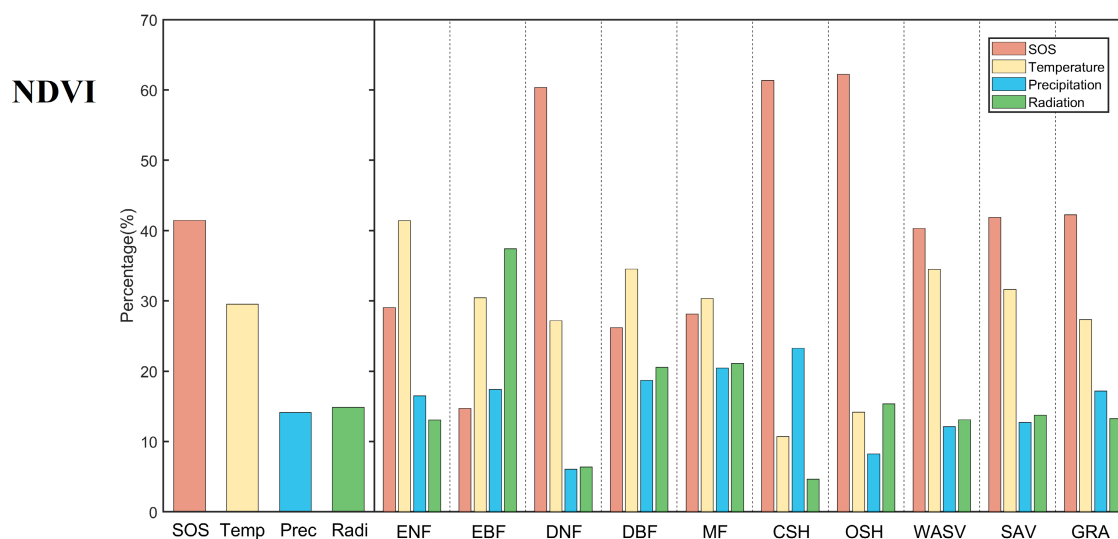


FIGURE 5 Contribution proportions of each driver (SOS, temperature, precipitation, radiation) to (a) POS_{CSIF} variations and (b) POS_{NDVI} variations over the Northern Hemisphere (> 30°N) and across biomes (averaged from all pixels in each biome) during 2001 to 2020. Subscripts CSIF and NDVI denoted the data sources. CSIF, contiguous solar-induced fluorescence; NDVI, normalized difference vegetation index; POS, peak of growing season; SOS, start of growing season. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

first hypothesis, this different ecological memory has been found between shrub and bunchgrass in previous study (Ogle et al., 2015). Theoretically, a great sensitivity of POS to SOS for biomes has large ecological memory. We therefore calculated variant driver sensitivities of POS and found the biggest SOS sensitivity of POS were in shrublands (Figure S7). For the second hypothesis, many studies have found the low climatic sensitivity in shrublands for spring phenology. For instance, Li et al. (2021) and Shen et al. (2014) found that spring phenology of shrublands and grasslands in the Northern Hemisphere exhibited lower climate sensitivity compared with other vegetation types (Li et al., 2021; Shen et al., 2014). In the current study, we additionally calculated the climatic sensitivity

of POS and found the lowest temperature sensitivity appeared in shrublands (Figure S7). This low climatic sensitivity could be related to the unique “fertile island” effect of shrublands. As shrubs affect the spatial distribution and circulation of nutrients by changing microclimate and concentrating organic matter under its canopy, the accumulation of soil resources under the canopy makes the shrub more resistant to disturbance (Pugnaire et al., 1996).

The strong positive correlations between POS and SOS in most areas of the Northern Hemisphere extratropical terrestrial ecosystems may include several mechanisms. First, the programmed apoptosis mechanism of cells and leaf structural constraints on longevity plays a direct effect (Lam, 2004; Reich et al., 1992). Trends in

POS are related to plant life cycle controlled by genotype and self-regulation mechanism (Yang et al., 2019). Second, earlier SOS leads to increase in sugar synthesis by photosynthesis, which stimulate cell division and leaf development in spring, thus fueling summer photosynthesis through biological carryover effect (Lian et al., 2021; Paul & Pellny, 2003; Piao et al., 2020). Increased photosynthetic activity accelerates plant seasonal cycle (Paul & Foyer, 2001). Third, an advanced SOS may lead to decreased soil water content through increasing transpiration. Plants may advance POS to maximize growth and access resources in competition (Lian et al., 2020; Radville et al., 2016). Therefore, the positive correlations between POS and SOS not only indicated the interaction between phenological events in response to climate change, but also included the indirect effects of phenology on environmental factors (Piao et al., 2019).

Nonetheless, we should also be aware of the non-negligible role of climatic factors. Our study confirmed that epidemic warming promoted advanced POS in most of the Northern Hemisphere and among biomes (Figure 5; Figure S4). Xu et al. (2016) found that warming-induced accumulation of growing degree days was response for advancing POS (Xu et al., 2016). Yang et al. (2019) found that an earlier POS may be attributed to warming-induced soil-water deficit (Yang et al., 2019). Specifically, in the relatively high latitudes of the Northern Hemisphere, temperature rather than other environmental factors played a more vital role in POS changes (Figure S6e,j). In addition, we found that light condition primarily controlled POS trends in eastern China and eastern United States, which was consistent with previous study in local region (Y. Liu et al., 2021). Limited radiation usually caused by insufficient sunshine intensity and hindered resource utilization of vegetation by affecting light-harvesting reaction of photosynthesis, and further impact vegetation growth and development (Gonsamo et al., 2015; Zhang et al., 2020).

5 | CONCLUSION

POS, as a transitional stage from greenup to senescence, was tightly connected to SOS and EOS. No matter from the physiological (tracked by SIF) or canopy (tracked by NDVI) level, we found an advanced POS in more than half of the Northern Hemisphere while reverse trends in northeast North America from 2001 to 2020. A partial derivative-based contributions analysis found these variations were mainly driven by SOS rather than climatic factors both at hemisphere and biome scales. This dominant effect of SOS was stronger in Shrublands (CSH and OSH) than in other biomes. Moreover, the effect of temperature on POS trends was greater than that of precipitation and radiation for most biomes. Our results suggested that the interaction between phenological events and the biome-specific POS should be considered in exploring global carbon cycle and phenological changes under climate change.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

All data that support the findings of this study are available from Dryad at 10.5061/dryad.4j0zpc8hn.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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